

The transported Jacobian sign has zero weak gradient

MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign

theorem

Uses signed transfer and weak Piola with the same test field to obtain the weak equation on the target.

Statement

Let $n \in \mathbb{N}_0$ and let $E = \mathbb{R}^n$ have the sup norm $\|\cdot\|_\infty$. Let $S, T \subseteq E$ be open, and let $f, g : E \rightarrow E$ be globally Lipschitz maps whose restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are mutually inverse. Fix a number $\ell > 0$ such that

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \quad (x, x' \in E).$$

Write $Df(x) : E \rightarrow E$ for the Fréchet derivative of f at x when it exists, and set $Df(x) = 0$ at other points. Define the signed Jacobian by $J_f(x) = \det Df(x)$.

Define the source sign $s_f : E \rightarrow \{-1, 1\}$ and its transport $\sigma_f : E \rightarrow \{-1, 1\}$ by

$$s_f(x) = \begin{cases} 1, & J_f(x) \geq 0, \\ -1, & J_f(x) < 0, \end{cases} \quad \sigma_f(y) = s_f(g(y)).$$

In particular, a zero determinant gives sign $+1$, and $|\sigma_f(y)| = 1$ for every y .

Let $W : E \rightarrow E$ be a C^1 vector field that vanishes outside a compact set $K \subseteq T$. Its divergence is $\operatorname{div} W(y) = \operatorname{tr} DW(y)$. With λ denoting Lebesgue measure,

$$\int_T \sigma_f(y) \operatorname{div} W(y) d\lambda(y) = 0.$$

The assertion holds for every such W and K .

Assumptions

The test field is C^1 on all of E , and its chosen compact set satisfies $K \subseteq T$ and $W = 0$ on $E \setminus K$. Neither connectedness nor finite measure of T is assumed.

The sets $S, T \subseteq E$ are open. The inverse and image conditions are

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T). \end{aligned}$$

The global bounds use specified constants $C_f, C_g \geq 0$ and $\ell > 0$:

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

The estimates hold on all of E , not just on S or T . The inverse identities are required only on S and T .

Conclusion

The scalar function σ_f has zero weak gradient on T : its product with the divergence of every admissible test field has integral zero. This is the integral formulation of a weak equation for a scalar function, not the divergence of a vector-valued sign.

Compactness is imposed on a set outside which the test field vanishes, not on the open set T . The statement itself does not require or assert that the sign is constant.

Proof route

The divergence of the test field is integrable. Substitute this function into signed transfer; the resulting source integral is zero by weak Piola.

▼ Proof steps

1. **Verify integrability of the observable.** Put $\phi(y) = \operatorname{div} W(y)$. Then

$$\phi(y) = 0 \quad (y \notin K), \quad \int_E |\phi(y)| d\lambda(y) = \int_K |\operatorname{div} W(y)| d\lambda(y) < \infty.$$

Indeed, DW is continuous because W is C^1 , and it vanishes off K because W vanishes locally there. Its trace is therefore continuous and zero off the compact set K . This is integrability of the compact-field divergence, applied to this same W and K . In particular, $\phi \in L^1(T, \lambda)$.

2. **Insert the observable into the two integral identities.** The complete substitution is

$$\begin{aligned} \int_T \sigma_f(y) \operatorname{div} W(y) d\lambda(y) &= \int_T \phi(y) \sigma_f(y) d\lambda(y) \\ &= \int_S \phi(f(x)) J_f(x) d\lambda(x) \\ &= \int_S (\operatorname{div} W)(f(x)) J_f(x) d\lambda(x) = 0. \end{aligned}$$

The second line is signed transfer with the maps f, g , the sets S, T , and the observable $\phi = \operatorname{div} W$; Step 1 supplies its integrability hypothesis. The last equality is weak Piola for the same f, g, W and the inclusion $K \subseteq T$. Since W and K were arbitrary, the weak equation follows.

Main citations

- Exact declaration and its source — `MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign`
- Signed transfer with the divergence observable — `MathlibAnnex.BilipschitzOrientation.signed_area_transfer`
- Weak Piola with the same field and carrier — `MathlibAnnex.BilipschitzOrientation.weak_piola`
- Integrability of the divergence of a compact C^1 field — `MathlibAnnex.BilipschitzOrientation.integrable_divergence`

- Total Jacobian sign, including the zero convention —
`MathlibAnnex.BilipschitzOrientation.domainJacobianSign`
- Transport of the sign by the specified inverse —
`MathlibAnnex.BilipschitzOrientation.targetJacobianSign`

Lean source signature (exact)

```
theorem weakDivergenceZero_targetJacobianSign {n : ℕ}
  (D : BilipschitzOpenData n) :
  WeakDivergenceZero volume D.target (targetJacobianSign D)
```

In the source	Mathematical meaning
(D : BilipschitzOpenData n)	The record described in this Card: open $S, T \subseteq E = \mathbb{R}^n$, maps $f, g : E \rightarrow E$ restricting to inverse bijections $S \leftrightarrow T$, global upper Lipschitz bounds C_f, C_g , and global lower bound $\ell \ x - x'\ _\infty \leq \ f(x) - f(x')\ _\infty$ with $\ell > 0$. The source name D is not the derivative symbol.
D.source; D.target; D.f; D.g	The local mathematical objects S, T, f, g , respectively, throughout this Card.

In the source	Mathematical meaning
targetJacobianSign D	The scalar function $\sigma_f(y) = s_f(g(y))$, where $J_f(x) = \det Df(x)$ and $s_f(x) = 1$ for $J_f(x) \geq 0$, $s_f(x) = -1$ for $J_f(x) < 0$. Zero has sign $+1$ under the cited definitions.
WeakDivergenceZero volume D.target (targetJacobianSign D)	The complete conclusion is the scalar predicate $\int_T \sigma_f(y) \operatorname{div} W(y) d\lambda(y) = 0$ for every global C^1 test field with a compact carrier $K \subseteq T$. volume is λ . It does not take an ordinary divergence of σ_f .

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 19–25:

```
noncomputable section

open Set MeasureTheory Filter
open scoped BigOperators ENNReal NNReal Topology

namespace MathlibAnnex
namespace BilipschitzOrientation
```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign`
Accepted content SHA-256: `4c3f7d4ed1c2ead75b23d6380f07535eb5fc21fb74de4b8b7e4f913aee4411d8`
Accepted source guide SHA-256: `64c0658743eb29f824c419c2cc8ad85c03e2fa02e241a7b69b3e1b87a6c4519c`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`